Late Breaking Results: Warpage-Aware Generative Floorplanning for Reliable Advanced Packaging

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***Abstract*—This paper presents the first warpage-aware generative learning-based floorplanning algorithm to effectively model the warpage effect and optimize the die floorplan on a fixed outlined substrate. With more heterogeneous materials and dense interconnects in advanced packaging, warpage is a main reliability concern and may degrade system performance. We present a novel transformer-based encoding scheme to learn node and edge representations, followed by parallel decoding and warpage-aware legalization to jointly minimize die displacement and warpage. Experimental results show that our algorithm improves warpage by 9.9% and wirelength by 8.3% on average, compared with the state-of-the-art work.**

I. Introduction

Warpage is an undesirable deformation caused by mismatched coefficients of thermal expansion (CTE) in a package, harming structural integrity and reliability, particularly in heterogeneous integration. Warpage can be calculated by the difference between the maximum and minimum points on the package's deformation surface, usually in the z-direction. Complex packaging structures like multiple stacked dies with diverse form factors, localized thermal concentrations from dense interconnects, and size problems like larger packages in heterogeneous integration may further amplify warpage.

A crucial technique to improve the warpage of packages is to design an optimized die floorplan, which necessitates an efficient, accurate warpage model for early-stage design planning. Recent work developed a *simplified* warpage model from Suhir's theory [1], suitable for efficient warpage optimization. Hsu *et al.* [2] presented the first warpage-aware floorplanning algorithm to optimize wirelength and warpage by calculating their in-die and out-of-die warpage cost on the transitive closure graph (TCG)-based floorplan representation. Chen *et al.* [3] used the same warpage model and employed mathematical programming to address multi-package co-design problems, optimizing for warpage and other production costs. However, the *simplified* model treats warpage as a local effect by calculating the effective material of each fixed partitioned grid from the package and ignoring a higher-order correction term. Further, their warpage model does not consider the special warpage periphery effect and is formulated only for separate dies instead of a continuous floorplan solution space. The warpage periphery effect, discovered empirically [4], suggests that placing dies near the substrate boundary can reduce warpage, emphasizing the need for a more accurate model to address this phenomenon in warpage modeling and optimization. To remedy these drawbacks, we propose the first machine learning (ML)-based multi-die package floorplanner enabled with our warpage model, which is more accurate than the *simplified* model and suitable for ML loss calculation. The floorplanner has two main features: (1) a generative learning architecture consisting of a transformer-based encoder, a physical-informed parallel decoder, and (2) a warpage-aware post-floorplanning algorithm to improve warpage and wirelength under minimum displacement from the ML model.

II. Problem Formulation

Given a package structure, a fixed substrate outline, a set of dies, I/O connections, and design rules, generate a floorplan with minimized warpage and wirelength without design rule violations.

III. Algorithms

Our floorplanning algorithm is an end-to-end model using gradient descent on our designed loss functions calculated on graphs to minimize warpage and wirelength. Figure 2 shows our algorithm flow, where the four stages are detailed in the following sections.

A. Feature Engineering

This stage transforms the package structure, die characteristics, and netlist connections into a graph, where nodes represent dies and edges represent connections. The node features summarize floorplan characteristics, including physical information, warpage-related material coefficients, and netlist connectivity. The edge features contain edge weights and eigenvector-derived values on the graph laplacian matrix, acknowledging the dense connections of typical packaging problems where die interconnections outweigh the edge density of a typical graph.

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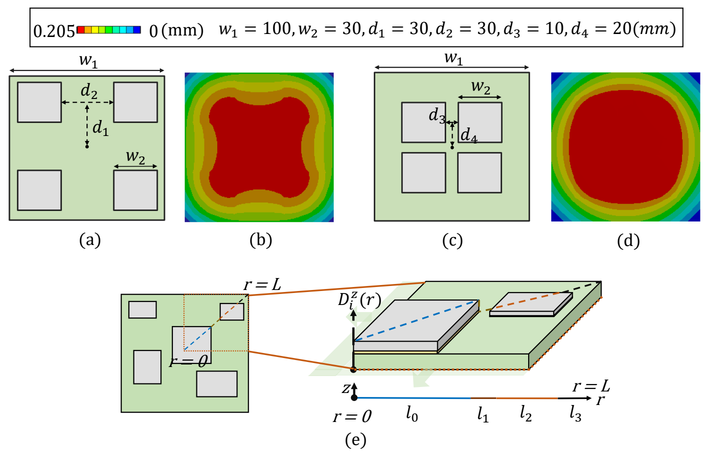

Fig. 1. (a) ∼ (d) The warpage periphery effect showing a 32% reduction in warpage for peripheral floorplan compared to a centric one (Note that all values are shifted to make the minimum equal to zero). (e) Our warpage calculation method.

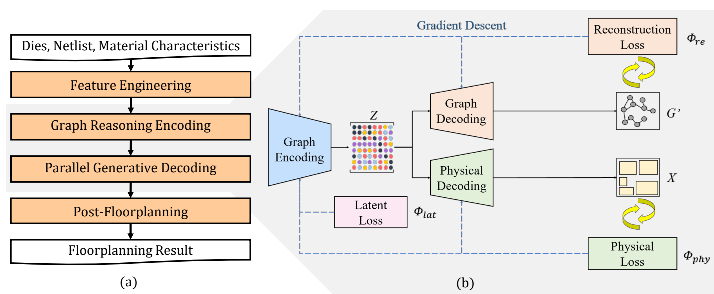

Fig. 2. (a) Our algorithm flow. (b) Graph encoding and parallel decoding for end-to-end multi-objective training.

B. Graph Reasoning Encoding

Our encoding architecture employs a Graph Transformer [5] to process both node and edge features. We integrate graph positional encoding using Laplacian eigenvectors to preserve structural information, capturing the k smallest non-trivial eigenvectors for improved representation. The latent space embedding, derived from the final transformer layer, is trained via unsupervised learning with a similarity loss function that clusters connected nodes while separating unconnected ones.

C. Parallel Generative Decoding

Our decoding architecture consists of two parallel components: a physical decoder that maps latent representations to two-dimensional coordinates and a graph decoder that reconstructs graph structures to refine node embeddings. The loss function of the decoders, Φ_{phy} and Φ_{re}, combines these objectives with weighted coefficients c, guiding optimization toward a high-quality floorplan with minimal displacement and improved structural integrity.

$$\Phi = \Phi_{phy} + \Phi_{re} = c_{wl}L_{wl} + c_{wp}L_{wp} + c_o L_o + c_b L_b + c_a L_a + c_g L_g + c_n L_n, \quad (1)$$

where L_{wl}, L_{wp}, L_o, L_b, L_a, L_g, and L_n are the wirelength loss, warpage loss, overlap loss, boundary loss, aspect ratio of the bounding box loss, graph structure loss, node embedding loss, respectively.

Our warpage model extends Tsai's [6] polynomial model by iteratively calculating assumed curvature from the package center. We partition the package into several sectors using radial axes with fixed-angle rotation, as shown in Figure 1(e). The segments S are separated by the intersection of a radial axis and dies, numbered from 0_{th} to L_{th}, where L is the segment abutting the substrate boundary. (L is 3 in the case of Figure 1(e)). We could calculate the radial deformation δ of these segments, formulated as follows:

$$\begin{aligned} \delta_i^z(r) &= \delta_{i-1}^z(l_{i-1}) + \left(\sum_{j=0}^{i-1} k_j(l_j - l_{j-1})\right) \cdot (r - l_{i-1}) \\ &+ \frac{1}{2}k_i(r-l_{i-1})^2, \quad i = 0 \sim L, \quad k_i = \frac{t\Delta\alpha\Delta T}{2\lambda D}, \end{aligned} \tag{2}$$

$$L_{wp} = \max_{i \in S} \delta_i^z(r) - \min_{i \in S} \delta_i^z(r), \tag{3}$$

where r is the distance from the substrate center, l is the distance of the former segment to the substrate center, k represents the effective curvature coefficient, i is the segment index, and t is the segment thickness. Other symbols including α, T, λ, and D have the same definitions in [6]. We define warpage loss as Equation (3), which is differentiable for gradient descent. To enhance the accuracy of warpage calculations and align the floorplanning result more closely with the legalized result, it is crucial to minimize die overlap. The overlap loss is defined as follows:

$$\begin{aligned} L_o &= \sum_{i,j \in V} A(i,j)\, P(i,j), \\ P(i,j) &= 1 + p\left(1 - \frac{|x_i - x_j| + |y_i - y_j|}{\frac{1}{2}(W_i + W_j + H_i + H_j)}\right), \end{aligned} \tag{4}$$

where $A(i,j)$ is the overlapping area of dies i and j, $P(i,j)$ is a penalty term used to avoid the complete overlap of two dies (with the same center coordinates), and p is a user-defined parameter controlling overlap penalty when the distance is close.

D. Post-Floorplanning

After ML floorplanning, some dies may overlap or extend beyond substrate boundaries. We employ a two-stage legalization scheme using constraint graphs and integer linear programming (ILP) to ensure legality with minimal displacement and optimized warpage. Constraint graphs capture the topological relationships between dies, with horizontal and vertical subgraphs encoding adjustment allowances. The ILP formulation first optimizes wirelength and overlap through 90-degree die rotations, defined as follows:

$$\begin{aligned} \min \sum_{u,v \in V} & (1-r_u)(1-r_v)\cdot\phi(u,v) + (1-r_u)r_v\cdot\phi(u,v_r) \\ &+ r_u(1-r_v)\cdot\phi(u_r,v) + r_u r_v\cdot\phi(u_r,v_r), \end{aligned} \tag{5}$$

where ϕ represents the weighted sum of wirelength and overlapping areas, and r_u, r_v, indicate the orientation of die u and v, respectively, which can be either vertical (90 degrees) or horizontal (0 degrees).

Then, we designed a minimum-displacement macro legalization algorithm to adjust die positions to satisfy the design rules (dies have required spacing and no overlaps) while minimizing the warpage periphery effect, formulated as follows:

$$\min \sum_{i \in V} \lambda_i^x \cdot d_i^x + \lambda_i^y \cdot d_i^y - \gamma_i \cdot (|x_i - x_c| + |y_i - y_c|), \tag{6}$$

where d_i^x and d_i^y are the x- and y-directional displacement of die i, λ_i^x and λ_i^y are positive weights that are proportional to the die size and connectivity density, x_c and y_c are the x- and y-coordinate of the substrate center, $|x_i - x_c|$ and $|y_i - y_c|$ are the periphery-induced terms that encourage die to be placed close to the boundary, γ_i controls the influence of the periphery-induced term, decided by k_i in Equation (2). Since both Equation (5) and Equation (6) are non-linear, we relax the orientation variable and absolute value terms to ensure solvability via an ILP solver. If the solution is infeasible, the algorithm reverts and selects an alternative candidate from the ML floorplanning stage.

IV. Experimental Results

We implemented our floorplanning algorithm in Python on a Linux workstation equipped with a 16-core AMD EPYC 7313 processor, NVIDIA RTX A6000 GPU, and 192GB memory. The ILP and finite element method (FEM)-based simulation were solved with the Gurobi Optimizer [7] and Ansys Mechanical [8], respectively. The simulation environment is set under uniform heat treatment during the manufacturing phase, presenting a more critical challenge than the operational phase. Our benchmarks include Hsu's (Warpage1∼Warpage5) and our enhanced cases (Warpage6∼Warpage10). Our enhanced cases were modified from the industrial ones, focusing on advanced packaging designs with uneven pin distributions and imbalanced neighboring connections. Across these benchmarks, die thickness ranged from 0.1 to 0.3 mm with CTE values between 3 and 10 $\frac{\mu m}{m \cdot K}$, while the substrate maintained a uniform 0.5 mm thickness with a CTE of 2.6 $\frac{\mu m}{m \cdot K}$ for polysilicon.

TABLE I
Comparison of the warpage, wirelength, and runtime for Hsu's and our design flow.

Case	Warpage (mm)		Wirelength (mm)		Runtime (s)	
	Hsu's	Ours	Hsu's	Ours	Hsu's	Ours
Warpage1	7.02	6.99	30934	26388	0.31	25.43
Warpage2	14.69	14.38	42545	41255	0.21	28.21
Warpage3	9.28	8.97	72092	63250	0.21	32.63
Warpage4	11.75	10.96	85366	91537	7.88	35.72
Warpage5	14.08	14.08	156929	138838	1.28	67.50
Warpage6	34.31	29.85	154095	140023	0.48	25.20
Warpage7	10.96	10.18	70674	69203	0.30	30.64
Warpage8	9.49	7.68	166861	134124	6.40	35.62
Warpage9	24.06	22.48	410195	409266	0.52	67.29
Warpage10	34.90	22.08	1033410	954377	1.26	153.20
Comp.	1.11	1	1.09	1	0.05	1

To validate the accuracy of our warpage model, we compared our results with Suhir's theory and Ansys simulations under two package structures: one centric die and two diagonal dies. For fair comparisons, the experiment parameters are the same as those in [6]. Quantitatively, our model reduces deformation deviation by 50.9% and 53.2% for the one-die and two-die cases, respectively, while improving warpage prediction by 13.6% and 29.1% compared to Ansys results. Qualitatively, it better aligns with simulation results, accurately capturing the dip between two dies, which previous models failed to identify. Our model and the *simplified* model compute warpage in under a second, significantly outperforming FEM-based simulations in runtime. With reasonable accuracy, our approach is well-suited for multi-die warpage optimization.

To evaluate the effectiveness of our proposed method, we compared our results with the state-of-the-art warpage floorplanner, Hsu's [2]. Leveraging generative learning, we employed transfer learning by pretraining a model on all test cases except the target design and fine-tuning it for each case. The training incorporated original and augmented data—using topological augmentation techniques such as random edge addition, edge masking, and edge weight scaling—with a fixed initial learning rate during pretraining. In the fine-tuning phase, the encoder weights were frozen to demonstrate its generality, while only the parallel decoder weights were adjusted. Pretraining required approximately 90 minutes across all benchmarks, with fine-tuning times detailed in Table I. Our results show that, on average, our method reduces warpage by 9.9% and wirelength by 8.3% compared to Hsu's approach, albeit with a 20.4-fold increase in runtime, demonstrating better performance over traditional TCG-based methods.

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